

Pengcheng Liao

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Summary

Quantum information researcher with expertise in quantum error correction, variational and machine-learning based quantum algorithms, and quantum state/process tomography.

Technical Skills

- **Quantum information:** quantum error correction, stabilizer and graph states, bosonic encodings, non-Gaussianity, quantum tomography
- **Algorithms:** variational quantum algorithms, quantum machine learning, complexity analysis
- **Numerical & software:** Python, QuTiP, PyTorch; large-scale numerical simulation and data analysis

Education

University of Southern California	Los Angeles, USA	<i>Jan 2022–Present</i>
<i>PhD in Electrophysics (expected 2026) Advisor: Prof. Quntao Zhuang</i>		
University of Calgary	Calgary, Canada	<i>Sep 2019–Dec 2021</i>
<i>MSc in Physics Advisors: Prof. David Feder & Prof. Barry Sanders</i>		
University of Chinese Academy of Sciences	Beijing, China	<i>Sep 2015–May 2019</i>
<i>BSc in Physics</i>		

Publications

- [1] **Pengcheng Liao**, Haowei Shi, Quntao Zhuang. *Variational Quantum Transduction*. arXiv:2603.03642 (2026).
- [2] Xiaobin Zhao, **Pengcheng Liao**, Francesco Anna Mele, Ulysse Chabaud, Quntao Zhuang. *Complexity of quantum tomography from genuine non-Gaussian entanglement*. Nature Communications (2025).
- [3] **Pengcheng Liao**, Bingzhi Zhang, Quntao Zhuang. *Quantum-enhanced learning with a controllable bosonic variational sensor network*. Quantum Science and Technology 9, 045040 (2024).
- [4] **Pengcheng Liao**, Quntao Zhuang. *Noisy entanglement testing for ranging and communication*. Physical Review Applied 22, 024007.
- [5] **Pengcheng Liao**, Barry Sanders, Tim Byrnes. *Quadratic speedup for quantum perceptron training*. Physical Review A 110, 062412.
- [6] **Pengcheng Liao**, Barry Sanders, David L. Feder. *Topological graph states and quantum error-correction codes*. Physical Review A 105, 042418.
- [7] **Pengcheng Liao**, David L. Feder. *Graph-state representation of the toric code*. Physical Review A 104, 012432.

Conference

- 2026, APS Global Physics Summit — *Variational Quantum Transduction*
- 2025, Optica Sensing Congress — *Noisy Entanglement Testing for Ranging and Communication*
- 2024, APS March Meeting — *Quantum-enhanced physical-layer data learning with a variational*

sensor network

- 2022, APS March Meeting — *Topological graph states and quantum error correction codes*
- 2021, The Fields Institute Workshop on Algebraic Graph Theory and Quantum Information — *Topological graph states*
- 2021, APS March Meeting — *Quantum circuits for topological state generation*